

Homework for Chap 14

Figure 1 shows absorption spectra of GaAs bulk material close to the bandgap measured at different temperatures.

1. Using an equation, explain why the absorption show redshift with increasing temperature.
2. Remarkable absorption enhancement close the bandgap is observed at the temperatures $T = 21$ K, 90 K, and 186 K. However, no obvious absorption enhancement is observed at $T = 294$ K. Explain the reason.

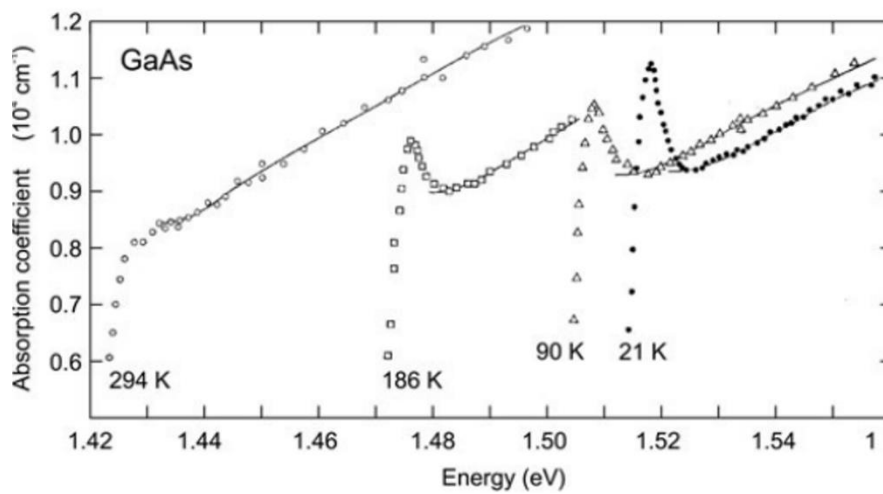


Figure 1. Absorption spectra of GaAs bulk material close to the bandgap at different temperatures.